ECM Fault Indication Not Indicated

Symptoms

- The DCU410 or remote panel does **not** indicate faults because it does **not** have power +24-VDC.
- The DCU410 unit and remote panel LEDs illuminated.
- The ECM has active faults.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

The SAE J1939 data link delivers alarm information, generated by the ECM, to the customer interface box. The customer interface box transfers alarm information to the DCU410 and remote panel.

This tree addresses the DCU410 power supply and return.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box.	
	STEP 1A. Check the DCU410 unit display for faults.	
	STEP 1A-1. Check the DCU410 power supply wire for voltage +24-VDC.	
	STEP 1B. Check the remote panel wiring.	
	STEP 1B-1. Check the remote panel power supply wire for voltage +24-VDC.	

STEPS		SPECIFICATIONS
STEP 2.	Check the customer interface box wiring.	
	STEP 2A. Check the DCU410 common alarm (common) wire for an open.	
	STEP 2A-1. Check the DCU410 common alarm (normally closed) wire for an open.	
	STEP 2B. Check the DCU410 common alarm (common) wire for a wire-to-wire short.	
	STEP 2B-1. Check the DCU410 common alarm (common) wire for a wire-to-wire short.	
	STEP 2B-2. Check the DCU410 common alarm (common) wire for a wire-to-wire short.	
	STEP 2C. Check the DCU410 common alarm (common) wire for a short to ground.	
	STEP 2C-1. Check the DCU410 common alarm (normally closed) wire for a short to ground.	
	STEP 2C-2. Check the DCU410 common alarm (normally closed) wire for a short to ground.	
	STEP 2D. Check the SAE J1939 data link supply and return wires for an open.	
	STEP 2D-1. Check the SAE J1939 data link supply and return wires for a wire-to-wire short.	
	STEP 2D-2. Check the SAE J1939 data link supply wire for short to ground.	

STEPS		SPECIFICATIONS
STEP 3.	Check the OEM wiring harness.	
	STEP 3A. Check the SAE J1939 data link supply and return wires for an open.	
	STEP 3A-1. Check the SAE J1939 data link supply and return wires for a wire-to-wire short.	
	STEP 3A-2. Check the SAE J1939 data link supply wire for a short to ground.	

STEP 1. Check the customer interface box.

STEP 1A. Check the DCU410 unit display for faults.

Conditions :

- Locate the DCU410 unit display.

Action	Specification/Repair	Next Step
Check the DCU410 unit display for indication of faults.	The DCU410 unit indicates fault(s)? **YES**	**1B**
	The DCU410 unit indicates fault(s)? **NO**	**1A-1**

STEP 1A-1. Check the DCU410 power supply wire for voltage +24-VDC.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step

Check the battery 1 voltage (switched power) at the DCU410 unit. • Place one test lead on the battery 1 voltage (switched power) supply wire at the DCU410 unit. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than +24-VDC? **YES** **Repair :** Check the batteries. Refer to the OEM service manual.	**Repair complete**
	Less than +24-VDC? **NO**	**1B**

STEP 1B. Check the remote panel wiring.

Conditions :
- Locate the remote panel display.

Action	Specification/Repair	Next Step
Check the remote panel display for an indication of faults.	The remote panel unit indicates fault(s)? **YES**	**Repair complete**
	The remote panel unit indicates fault(s)? **NO**	**1B-1**

STEP 1B-1. Check the remote panel power supply wire for voltage +24-VDC.

Conditions :
- Open the remote panel.

Action	Specification/Repair	Next Step

Check the battery 1 voltage (switched power) at the DCU410 unit. • Place one test lead on the battery 1 voltage (switched power) supply wire at the remote panel unit. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than +24-VDC? **YES** **Repair :** Check the batteries. Refer to OEM service manual.	**Repair complete**
	Less than +24-VDC? **NO**	**2A**

STEP 2. Check the customer interface box wiring.

STEP 2A. Check the DCU410 common alarm (common) wire for an open.

Conditions :

- Open the customer interface box.
- Disconnect the DCU410 common alarm (common) wire at the DCU410 and X4 connection.

Action	Specification/Repair	Next Step

<table>
<tr>
<td rowspan="2">Check the DCU410 common alarm (common) wire at the DCU410 unit and X4 connection for an open.
Place one test lead on the DCU410 common alarm (common) wire at the DCU410 unit.
Place the other test lead on the DCU410 common alarm (common) at the X4 connection.
Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</td>
<td>Less than 10 ohms?
YES</td>
<td>2A-1</td>
</tr>
<tr>
<td>Less than 10 ohms?
NO
Repair :
Replace the wire. Refer to the Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</td>
<td>Repair complete</td>
</tr>
</table>

STEP 2A-1. Check the DCU410 common alarm (normally closed) wire for an open.

<table>
<tr>
<td colspan="3">Conditions :
Open the customer interface box.
Disconnect the DCU410 common alarm (normally closed) wire at the DCU410 and X4 connection.</td>
</tr>
<tr>
<th>Action</th>
<th>Specification/Repair</th>
<th>Next Step</th>
</tr>
</table>

Check the DCU410 common alarm (normally closed) wire at the DCU410 and X4 connection for an open. • Place one test lead on the DCU410 common alarm (normally closed) wire at the DCU410 unit. • Place the other test lead on the DCU410 common alarm (normally closed) at the X4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**2B**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to the Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 2B. Check the DCU410 common alarm (common) wire for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the DCU410 common alarm (common) wire at the DCU410 unit and X4 connection.

Action	**Specification/Repair**	**Next Step**

Check the DCU410 common alarm (common) wire at the DCU410 unit and X4 connector for a wire-to-wire short. • Place one test lead on the DCU410 common alarm (common) wire at the DCU410 unit. • Place the other test lead on all other wires at the DCU410 unit. • Place one test lead on the DCU410 common alarm (common) wire at the X4 connection. • Place the other test lead on all other wires at the X4 connection. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to the Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**2B-1**

STEP 2B-1. Check the DCU410 common alarm (normally closed) wire for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the DCU410 common alarm (normally closed) wire at the DCU410 unit and X4 connection.

Action	Specification/Repair	Next Step

Check the DCU410 common alarm (normally closed) wire at the DCU410 unit and X4 connector for a wire-to-wire short. • Place one test lead on the DCU410 common alarm (normally closed) wire at the DCU410 unit. • Place the other test lead on all other wires at the DCU410 unit. • Place one test lead on the DCU410 common alarm (normally closed) wire at the X4 connection. • Place the other test lead on all other wires at the X4 connection. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to the Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**2B-2**

STEP 2B-2. Check the DCU410 common alarm (normally open) wire for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the DCU410 common alarm (normally open) wire at the DCU410 unit and X4 connection.

Action	Specification/Repair	Next Step

<table>
<tr>
<td rowspan="2">Check the DCU410 common alarm (normally open) wire at the DCU410 unit and X4 connector for a wire-to-wire short.
• Place one test lead on the DCU410 common alarm (normally open) wire at the DCU410 unit.
• Place the other test lead on all other wires at the DCU410 unit.
• Place one test lead on the DCU410 common alarm (normally open) wire at the X4 connection.
• Place the other test lead on all other wires are the X4 connection.
Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</td>
<td>Less than 10 ohms?
YES
Repair :
Replace the wire. Refer to the Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)
Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.</td>
<td>Repair complete</td>
</tr>
<tr>
<td>Less than 10 ohms?
NO</td>
<td>2C</td>
</tr>
</table>

STEP 2C. Check the DCU410 common alarm (common) wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the DCU410 common alarm (common) wire at the DCU410 unit and X4 connection.

Action	Specification/Repair	Next Step

Check the DCU410 common alarm (common) wire at the DCU410 unit and X4 connection for a short to ground. • Place one test lead on the DCU410 common alarm (common) wire at the DCU410 unit. • Place the other test lead on panel ground. • Place one test lead on the DCU410 common alarm (common) wire at the X4 connection. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to the Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html) Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	**Repair complete**
	Less than 10 ohms? **NO**	**2C-1**

STEP 2C-1. Check the DCU410 common alarm (normally closed) wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the DCU410 common alarm (normally closed) wire at the DCU410 unit and X4 connection.

Action	Specification/Repair	Next Step

Check the DCU410 common alarm (normally closed) wire at the DCU410 unit and X4 connector for a short to ground. • Place one test lead on the DCU410 common alarm (normally closed) wire at the DCU410 unit. • Place the other test lead on panel ground. • Place one test lead on the DCU410 common alarm (normally closed) wire at the X4 connection. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to the Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html) Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	**Repair complete**
	Less than 10 ohms? **NO**	**2C-2**

STEP 2C-2. Check the DCU410 common alarm (normally closed) wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the DCU410 common alarm (normally open) wire at the DCU410 unit and X4 connection.

Action	Specification/Repair	Next Step

Check the DCU410 common alarm (normally open) wire at the DCU410 unit and X4 connector for a short to ground. • Place one test lead on the DCU410 common alarm (normally open) wire at the DCU410 unit. • Place the other test lead on panel ground. • Place one test lead on the DCU410 common alarm (normally open) wire at the X4 connection. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to the Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html) Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	**Repair complete**
	Less than 10 ohms? **NO**	**2D**

STEP 2D. Check the SAE J1939 supply and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the SAE J1939 data link supply and return wires at the DCU410 unit and X4 connection.
- Disconnect the C3 connector.

Action	Specification/Repair	Next Step

Check the SAE J1939 data link supply and return wires at the DCU410 unit, X4 connection, and C3 connector for an open. • Place one test lead on the SAE J1939 data link supply (C3) wire at the DCU410 unit. • Place the other test lead on the SAE J1939 data link supply (C3) at the X4 connection. • Place one test lead on the SAE J1939 data link supply (C3) wire at the DCU410 unit. • Place the other test lead on the SAE J1939 data link supply (C3) at the C3 connector. • Place one test lead on the SAE J1939 data link return (C3) wire at the DCU410 unit. • Place the other test lead on the SAE J1939 data link return (C3) at the X4 connection. • Place one test lead on the SAE J1939 data link return (C3) wire at the DCU410 unit. • Place the other test lead on the SAE J1939 data link return (C3) at the C3 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**2D-1**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to the Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 2D-1. Check the SAE J1939 data link supply and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the SAE J1939 data link supply and return wires at the DCU410 unit and X4 connection.
- Disconnect the C3 connector.

Action	Specification/Repair	Next Step

Check the SAE J1939 data link supply and return wires at the DCU410 unit, X4 connection, and C3 connector for a wire-to-wire short. - Place one test lead on the SAE J1939 data link supply (C3) wire at the DCU410 unit. - Place the other test lead on all other wires at the DCU410 unit. - Place one test lead on the SAE J1939 data link return (C3) wire at the DCU410 unit. - Place the other test lead on all other wires at the DCU410 unit. - Place one test lead on the SAE J1939 data link supply (C3) wire at the X4 connector. - Place the other test lead on all other wires at the X4 connector. - Place one test lead on the SAE J1939 data link return (C3) wire at the X4 connector. - Place the other test lead on all other wires at the X4 connector. - Place one test lead on the SAE J1939 data link supply (C3) wire at the C3 connector. - Place the other test lead on all other wires at the C3 connector.	Less than 10 ohms? **YES** **Repair :** Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	**Repair complete**

• Place one test lead on the SAE J1939 data link return (C3) wire at the C3 connector. • Place the other test lead on all other wires at the C3 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **NO**	**2D-2**

STEP 2D-2. Check the SAE J1939 data link supply wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the SAE J1939 data link supply and return wires at the DCU410 unit and X4 connection.
- Disconnect the C3 connector.

Action	Specification/Repair	Next Step

Check the SAE J1939 data link supply wire at the DCU410 unit, X4 connection, and C3 connector for a short to ground. • Place one test lead on the SAE J1939 data link supply wire at the DCU410 unit. • Place the other test lead on panel ground. • Place one test lead on the SAE J1939 data link supply wire at the X4 connection. • Place the other test lead on panel ground. • Place one test lead on the SAE J1939 data link supply wire at the C3 connector. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	**Repair complete**
	Less than 10 ohms? **NO**	**3A**

STEP 3. Check the OEM wiring harness.

STEP 3A. Check the SAE J1939 data link supply and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the SAE J1939 data link supply and return wires at the X4 connection.

Action	Specification/Repair	Next Step

Check the SAE J1939 data link supply and return wires for an open. • Place one test lead on the SAE J1939 data link supply wire at the X4 connection. • Place the other test lead on the SAE J1939 data link supply wire at the SAE J1939 data link service port connector. • Place one test lead on the SAE J1939 data link return wire at the X4 connection. • Place the other test lead on the SAE J1939 data link return wire at the SAE J1939 data link service port connector.	Less than 10 ohms? **YES**	**3A-1**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to the OEM service manual.	**Repair complete**

STEP 3A-1. Check the SAE J1939 data link supply and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the SAE J1939 data link supply and return wires at the X4 connection.

Action	Specification/Repair	Next Step

Check the SAE J1939 data link supply and return wires at the X4 connection for a wire-to-wire short. • Place one test lead on the SAE J1939 data link supply wire at the X4 connection. • Place the other test lead on all other wires at the X4 connection. • Place one test lead on the SAE J1939 data link return wire at the X4 connection. • Place the other test lead on all other wires at the X4 connection. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**3A-2**
	Less than 10 ohms? **NO** **Repair :** Replace the SAE J1939 data link service port connector. Contact a Cummins® Authorized Repair Location.	**Repair complete**

STEP 3A-2. Check the SAE J1939 data link supply wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the SAE J1939 data link supply wire at the X4 connection and SAE J1939 data link service port connector.

Action	Specification/Repair	Next Step

Check the SAE J1939 data link supply wire for a short to ground. • Place one test lead on the SAE J1939 data link supply wire at the X4 connection. • Place the other test lead on engine ground. • Place one test lead on the SAE J1939 data link return wire at the SAE J1939 data link service port connector. • Place the other test lead on engine ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**Contact a Cummins® Authorized Repair Location.**
	Less than 10 ohms? **NO** **Repair :** Replace the SAE J1939 data link service port connector. Contact a Cummins® Authorized Repair Location.	**Repair complete**

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